

Name: \_\_\_\_\_

Class: \_\_\_\_\_

Date: \_\_\_\_\_

# Universal STEM Worksheet Engine

Secondary STEM Category | ISNSW Curriculum Conference 2026

Primary → Stage 6 · Pure TikZ, zero embedded images · NESAs-aligned



Scan or click:  
AI Agent Competition

One engine, one design language — from a Year 2 child measuring a friendly stick-figure with a ruler, to a Stage 6 student resolving forces, analysing AC circuits and proving roots of unity. Every diagram is generated in **pure TikZ**: no embedded images. Outcome codes are **indicative drafts for teacher verification** against the current NSW NESA syllabuses. **Note:** Document layout is highly compressed for ISNSW showcase purposes. In-production artifacts generate with stage-appropriate spacing to manage cognitive load.

- Primary · K–6
- Junior STEM · Yr 7–10
- Senior STEM · Yr 11–12
- Extension

PRIMARY

Early Years · Time & Measuring Length

Learning Intentions & Success Criteria

I can read analogue time, measure length in centimetres starting from 0, and compare heights using a scale.

Telling the Time

Y2

Analogue clock

[2 marks]

NESA: Measurement & Geometry

The clock shows a time in the morning. Write it in 12-hour and 24-hour formats.



12hr: \_\_\_\_\_ 24hr: \_\_\_\_\_

Using a Ruler

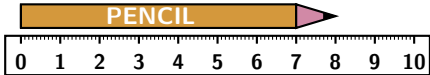
Y2

Measure pencil

[1 mark]

NESA: Measurement & Geometry

Read the ruler and state the exact length of the pencil in centimetres.



Length: \_\_\_\_\_ cm

Comparing Heights

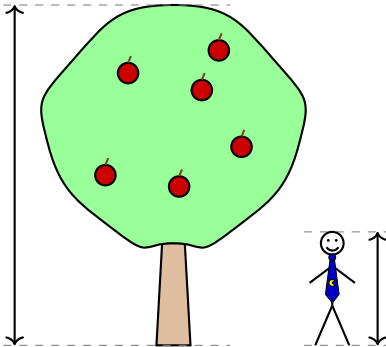
Y2

How tall?

[2 marks]

NESA: Measurement & Geometry

Use the scale to measure the **height** of the tree and Darra. How much taller is the tree?



Tree: \_\_\_\_\_ cm

Darra: \_\_\_\_\_ cm

Difference: \_\_\_\_\_ cm

**JUNIOR STEM**

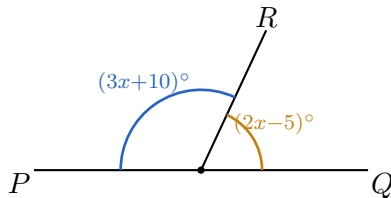
**Stage 4–5 · Geometry, Algebra, Trigonometry & Science**

**Angle Reasoning**

**Stage 4 Angles on a line [2 marks]**

NESA: Angle relationships

Line  $POQ$  is straight and ray  $OR$  meets it at  $O$ . Form an equation, solve for  $x$ , then state  $\angle POR$ .

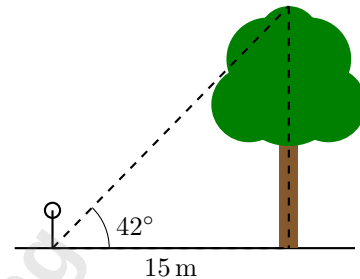


**Angle of Elevation**

**Stage 5 Trigonometry [2 marks]**

NESA: Trigonometry & Pythagoras

Standing 15 m from a tree, a clinometer reads  $42^\circ$  to the top. Find the height to 2 d.p.

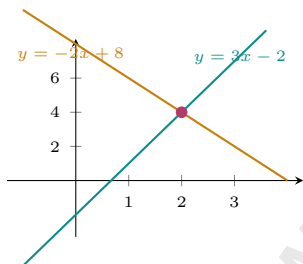


**Simultaneous Lines**

**Stage 4 Point of intersection [2 marks]**

NESA: Linear relationships

The lines  $y = 3x - 2$  and  $y = -2x + 8$  are graphed below. State their point of intersection.

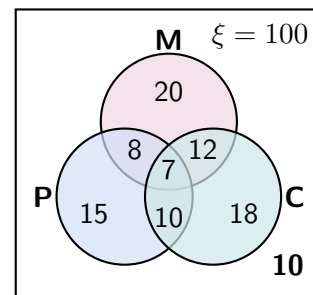


**Set Theory & Probability**

**Stage 5 Venn diagrams [3 marks]**

NESA: Probability / data

100 students study Maths (M), Physics (P) or Chemistry (C). Find  $P(P | M)$ .

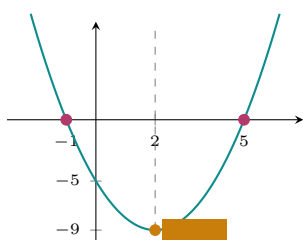


**The Parabola**

**Stage 5 Quadratic features [3 marks]**

NESA: Non-linear relationships

For  $y = x^2 - 4x - 5$ , state the  $x$ -intercepts, axis of symmetry, and the turning point.

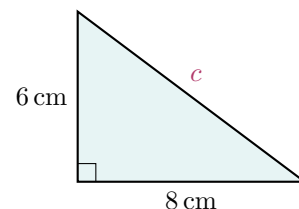


**Pythagoras**

**Stage 5 Find the hypotenuse [2 marks]**

NESA: Trigonometry & Pythagoras

A right-angled triangle has the two shorter sides shown. Find the length of the hypotenuse  $c$ .



$$c = \sqrt{a^2 + b^2}$$

## Investigation Skills

## Stage 5 Science — Pendulum Investigation

[4 marks]

A simple pendulum is released and you investigate how **string length** affects the **period** of one swing. State the independent and dependent variables, **two** controlled variables, a testable hypothesis, and **one** safety risk with its control. *NESA: Working Scientifically — questioning, predicting, planning*

IV: \_\_\_\_\_

Hypothesis / risk: \_\_\_\_\_

DV: \_\_\_\_\_

Controlled: \_\_\_\_\_, \_\_\_\_\_

## SENIOR STEM

## Stage 6 · Physics, Chemistry, Software &amp; Extension

## Learning Intentions &amp; Success Criteria

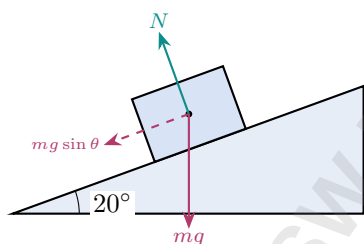
I can resolve forces on an incline, analyse a series RLC circuit at resonance, operate with 3D vectors, represent organic reactions, prove results about roots of unity, interpret a slope field, model a normal distribution, and reduce a logic circuit.

## Mechanics · Inclined Plane

## Physics Free-body diagram [3 marks]

*NESA: Physics — Dynamics*

A block of mass  $m = 6.0 \text{ kg}$  rests on a *frictionless* incline at  $\theta = 20^\circ$ . Draw the free-body diagram and find the acceleration.

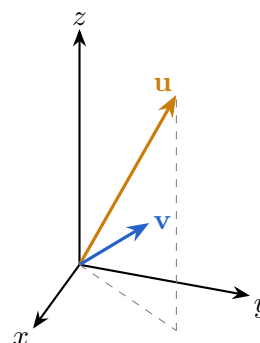


## Extension · 3D Vectors

## Maths Ext Dot product &amp; angle [3 marks]

*NESA: Enrichment — vectors in 3D*

For  $\mathbf{u} = 2\mathbf{i} + 3\mathbf{j} + 6\mathbf{k}$  and  $\mathbf{v} = \mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$ , find  $\mathbf{u} \cdot \mathbf{v}$  and the angle between them.

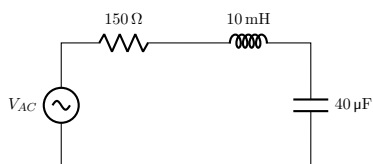


## Electromagnetism · RLC

## Physics Series resonance [3 marks]

*NESA: Physics — Electromagnetism*

For the series RLC circuit with  $R = 150 \Omega$ ,  $L = 10 \text{ mH}$  and  $C = 40 \mu\text{F}$ , calculate the resonant frequency in hertz.



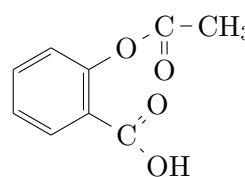
$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

## Chemistry · Organic

## Chem Functional groups &amp; synthesis [4 marks]

*NESA: Chemistry — Organic Chemistry*

Below is the skeletal structure of Aspirin. Identify the three primary functional groups present. Write the balanced chemical equation for the synthesis of Aspirin reacting salicylic acid with acetic anhydride.



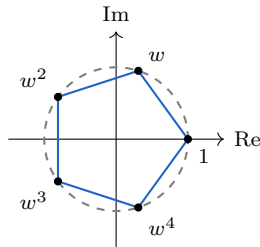
Equation: \_\_\_\_\_

### Extension · Complex Numbers

#### Ext 2 Roots of unity [3 marks]

NESA: Mathematics Extension 2

The diagram shows the 5th roots of unity, where  $w = \text{cis}\left(\frac{2\pi}{5}\right)$ . Prove  $1 + w + w^2 + w^3 + w^4 = 0$ .

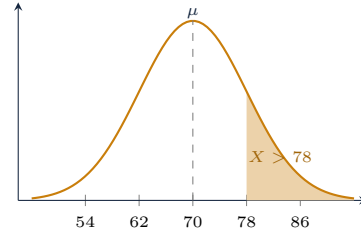


### Statistics · Normal Distribution

#### Maths Adv The empirical rule [2 marks]

NESA: Mathematics Advanced — statistical analysis

Test scores are normally distributed with mean  $\mu = 70$  and standard deviation  $\sigma = 8$ . Use the shaded region to estimate  $P(X > 78)$ .

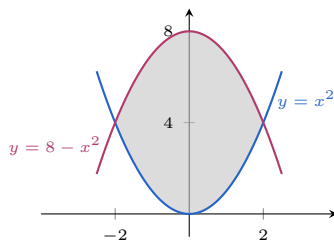


### Calculus · Area Between Curves

#### Maths Adv Definite integral [3 marks]

NESA: Mathematics Advanced

Find the exact area of the shaded region bounded by  $y = x^2$  and  $y = 8 - x^2$ .



### Multiple Choice · Logarithms

#### Maths Adv Logarithmic equations [1 mark]

NESA: Mathematics Advanced

What is the solution to  $\log_2(x) + \log_2(x - 3) = 2$ ?

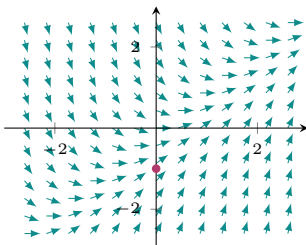
- A.  $x = -1$
- B.  $x = 4$
- C.  $x = 4$  and  $x = -1$
- D. No real solutions

### Calculus · Slope Field

#### Maths Ext Direction field [3 marks]

NESA: Mathematics Extension 1 — differential equations

The slope field for  $\frac{dy}{dx} = x - y$  is shown. Sketch the solution curve passing through  $(0, -1)$  and describe its long-term behaviour.

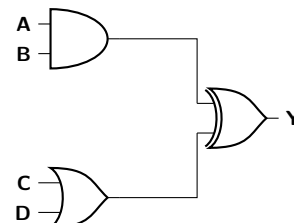


### Software & Engineering

#### Soft Design Boolean circuit [3 marks]

NESA: Software Engineering

Write the Boolean expression for the output  $Y$ , then state the value of  $Y$  when  $A = 1, B = 1, C = 0, D = 0$ .



**SHOWCASE** Scalable Visual Pedagogy & Cross-Curricular**Learning Intentions & Success Criteria**

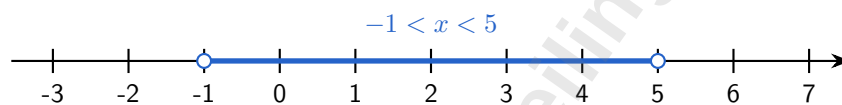
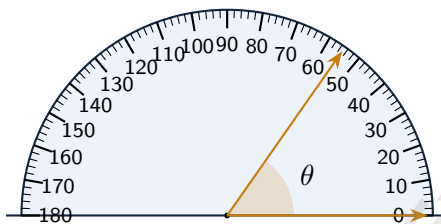
**Competition Note:** The following components demonstrate the engine's zero-shot ability to scale a single visual framework across distinct educational stages (Year 2 vs. Stage 6) and adapt to entirely different faculties (History, English) using pure TikZ.

**Component 1 · The Number Line — One Macro, Any Stage**

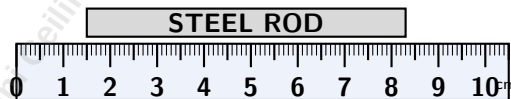
**Year 2:** Count the jumps to find where the kangaroo lands.



**Stage 6:** The *same* coordinate axis component now represents the solution of  $|x - 2| < 3$ .

**Component 2 · Visual Protractor**

Measure  $\theta$  to the nearest degree.

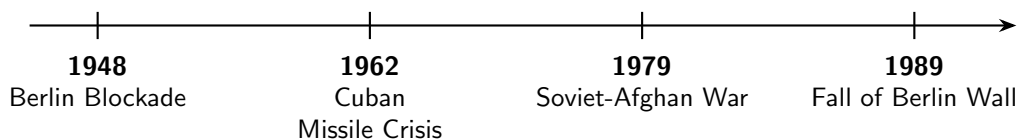
**Component 3 · Precision Ruler**

Read the rod's length to the nearest millimetre.

**Cross-Curricular · Historical Timeline****Modern History** Timeline Analysis**[4 marks]**

NESA: Modern History

The timeline below maps four critical flashpoints of the Cold War. Select ONE of these events and evaluate its long-term impact on United States-Soviet Union geopolitical relations.



## Cross-Curricular · Literary Analysis

**English Adv** Translation & Literary Devices**[4 marks]**

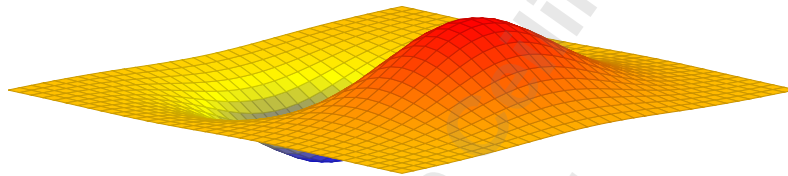
NESA: English Advanced

Translate the following excerpt from Shakespeare's *Hamlet* into modern Australian vernacular. Additionally, identify and analyse the primary literary device used in the final two lines to convey mental anguish.

**William Shakespeare, Hamlet**

To be, or not to be, that is the question:  
Whether 'tis nobler in the mind to suffer  
The slings and arrows of outrageous fortune,  
Or to take arms against a sea of troubles...

## Component 4 · Advanced 3D Topology &amp; Data Vis



**Advanced PGFPlots:** Render complex 3D topological surfaces and multivariable calculus functions natively without relying on external image rendering engines.

## Cross-Curricular · Business Studies

**Business** Strategic Management**[3 marks]**

NESA: Business Studies

A legacy automotive company has invested \$500 million into developing a highly efficient internal combustion engine. Before completion, a breakthrough in solid-state batteries makes electric vehicles significantly cheaper to produce. The board argues they must launch the new combustion engine to “recover our investment.” Identify the logical fallacy driving the board's argument and explain why proceeding could harm their long-term market position.

## AI Pedagogy Engines · Deep Learning Extensions

**Spot the Error** Calculus bounds **[2 marks]**

A student evaluates  $\int_0^2 2x\sqrt{x^2+1} dx$ .

**L1:** Let  $u = x^2 + 1$ ,  $du = 2x dx$ .

**L2:**  $= \int_0^2 \sqrt{u} du$

**L3:**  $= [\frac{2}{3}u^{3/2}]_0^2 = \frac{4\sqrt{2}}{3}$ .

**Error in Line:** \_\_\_\_\_

**Parameter Shift** Dynamics **[2 marks]**

Refer to the **Inclined Plane** question. If the angle  $\theta$  is increased to  $30^\circ$  and the mass doubled to 12.0 kg:

**a)** Does acceleration increase, decrease, or stay the same?

**b)** Does the normal force  $N$  increase or decrease?

ASSESS

Automated Marking System & Feedback

| TikZ-Typeset Marking Rubric — Working Mathematically / Scientifically |                                |                                |                                    |      |
|-----------------------------------------------------------------------|--------------------------------|--------------------------------|------------------------------------|------|
| Criterion                                                             | Developing (1)                 | Proficient (2)                 | Exemplary (3)                      | Mark |
| Method                                                                | Partially valid approach       | Selects a correct method       | Efficient, fully justified method  |      |
| Working                                                               | Steps unclear or gaps          | Logical steps, minor slips     | Complete, clearly sequenced        |      |
| Accuracy & units                                                      | Frequent errors; units missing | Mostly accurate; correct units | Accurate; units & sig figs correct |      |
| Communication                                                         | Hard to follow                 | Clear notation                 | Concise, syllabus-aligned notation |      |

| Mark Allocation & Syllabus Map |                                                                                          |       |
|--------------------------------|------------------------------------------------------------------------------------------|-------|
| Band                           | Focus areas sampled                                                                      | Marks |
| Primary (K–6)                  | Time, length, comparing height                                                           | 5     |
| Junior STEM (Yr 7–10)          | Angles, linear, quadratic, trig, probability, Pythagoras, investigation                  | 18    |
| Senior STEM (Yr 11–12)         | Dynamics, electromagnetism, organic chemistry, calculus, slope fields, statistics, logic | 22    |
| Extension                      | 3D vectors, roots of unity                                                               | 6     |
| Total: _____                   |                                                                                          | / 51  |

What Went Well

Even Better If

Overall: ☐ Developing ☐ Proficient ☐ Exemplary

Grade: \_\_\_\_\_

AI Tutor Diagnostic · Cognitive Profile & Adaptive Remediation

Algorithmic Insight (Mental Maths Mentor):

Student demonstrates high procedural fluency in algebraic manipulation and calculus, but exhibits cognitive overload when translating text to visual-spatial representations (e.g., 3D Vectors, Geometry).

Adaptive Next Steps:

- Decrease: Repetitive procedural algebra tasks.
- Increase: GeoGebra-linked spatial reasoning tasks.
- Mentorship Note: High resilience shown in multi-step problem solving. Acknowledge effort and pivot to visual conceptual scaffolding for the next module.

Comm.

Algebra

Calculus

Logic

Spatial

## ANSWERS

## Teacher Answer Key — Worked Reasoning

**Time.** 12-hour: 10:10 AM; 24-hour: 10:10.

**Ruler.** 7.0 cm (the pencil tip aligns with the 7 mark; insist the student starts at 0).

**Heights.** Read each double-arrow scale: tree  $\approx 6$  units, Darra  $\approx 2$  units; the tree is about 4 units taller (accept consistent measured values).

**Angles.**  $POQ$  straight  $\Rightarrow \angle POR + \angle ROQ = 180^\circ$ :  $(3x + 10) + (2x - 5) = 180 \Rightarrow 5x + 5 = 180 \Rightarrow x = 35$ . Hence  $\angle POR = 3(35) + 10 = 115^\circ$ .

**Lines.**  $3x - 2 = -2x + 8 \Rightarrow 5x = 10 \Rightarrow x = 2$ ,  $y = 4$ . Intersection (2, 4).

**Parabola.**  $x^2 - 4x - 5 = (x - 5)(x + 1)$ ; intercepts  $x = -1, 5$ ; axis  $x = 2$ ; turning point (2, -9).

**Elevation.**  $\tan(42^\circ) = \frac{h}{15} \Rightarrow h = 15 \tan(42^\circ) \approx 13.51$  m.

**Venn.** Only one subject:  $20 + 15 + 18 = 53$ .  $P(P | M) = \frac{n(P \cap M)}{n(M)} = \frac{8 + 7}{20 + 8 + 12 + 7} = \frac{15}{47}$ .

**Pythagoras.**  $c = \sqrt{8^2 + 6^2} = \sqrt{100} = 10$  cm.

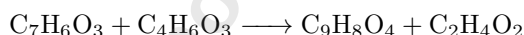
**Pendulum.** IV: string length; DV: period. Controlled (any two): bob mass, release angle, timing method, location. Hypothesis: longer string  $\Rightarrow$  longer period. Risk: bob striking a student — keep the path clear, limit amplitude.

**Inclined plane.** Frictionless:  $a = g \sin \theta = 9.8 \sin(20^\circ) \approx 3.35 \text{ m} \cdot \text{s}^{-2}$  down the plane ( $N = mg \cos \theta$  balances the perpendicular component).

**RLC.**  $f_r = \frac{1}{2\pi\sqrt{LC}}$ ,  $L = 10 \times 10^{-3}$ ,  $C = 40 \times 10^{-6}$ .  $\sqrt{LC} \approx 6.325 \times 10^{-4}$ , so  $f_r \approx 251.65$  Hz.

**3D vectors.**  $\mathbf{u} \cdot \mathbf{v} = (2)(1) + (3)(2) + (6)(2) = 20$ ;  $|\mathbf{u}| = 7$ ,  $|\mathbf{v}| = 3$ , so  $\cos \theta = \frac{20}{21} \Rightarrow \theta \approx 18^\circ$ .

**Chemistry.** Functional groups: carboxylic acid, ester, aromatic (benzene) ring. Synthesis:



(salicylic acid + acetic anhydride  $\rightarrow$  aspirin + acetic acid).

**Roots of unity.** The roots solve  $z^5 - 1 = 0$ . Their sum is the geometric series  $\frac{1 - w^5}{1 - w}$ ; since  $w^5 = 1$  the numerator is 0 (and  $w \neq 1$ ), so the sum is 0.

**Area between curves.** Intersect:  $x^2 = 8 - x^2 \Rightarrow x = \pm 2$ .  $A = \int_{-2}^2 [(8 - x^2) - x^2] dx = \int_{-2}^2 (8 - 2x^2) dx = \frac{64}{3}$  square units.

**Slope field.** Solution through (0, -1) rises, crosses below the line  $y = x - 1$ , and is asymptotic to  $y = x - 1$  as  $x \rightarrow \infty$  (the particular solution is  $y = x - 1$ ).

**Normal distribution.**  $z = \frac{78 - 70}{8} = 1$ . By the empirical rule  $P(X > 78) = P(Z > 1) \approx 0.16$  (16%).

**Logic.**  $Y = (A \wedge B) \oplus (C \vee D)$ . With  $A = B = 1$ ,  $C = D = 0$ :  $(1) \oplus (0) = 1$ .

**MC Logarithms.** B ( $x = 4$ ).  $x = -1$  is outside the log domain.

**Spot the Error.** Line 2. The integral bounds were not changed to  $u$ -values (should be  $u = 1$  to 5).

**Parameter Shift.** a) Increases ( $a = g \sin \theta$  is mass-independent, and  $\sin(30^\circ) > \sin(20^\circ)$ ). b) Increases (doubling the mass outweighs the slight drop in  $\cos \theta$ ).

**Business Studies.** Identifies the Sunk Cost Fallacy. Proceeding ignores future viability; EVs will undercut ICE vehicles, losing market share and compounding financial losses.



ISNSW AI Agent Competition



Try the Custom Gem!



Scan to view Agent Instructions

Name:

Date:

**My Measurement Challenge!***Learning to measure with cubes and rulers***Part 1: Match the Words**

Draw a line from the word to the matching picture.

String

Rubber

Tall



1. **Word Bank Challenge:** Use the words below to finish the sentence.

string   tall   rubbers

To find out how

my tower is, I can measure it with

**Part 2: Hands-On Non-Standard Units***Grab your measuring tools and let's get to work!*

2. **Measure Your Desk!**  
Place your hands flat on your desk, side-by-side. Measure how long your desk is from one side to the other.

My desk is                  hands long.

3. How many rubbers long is this shoe?

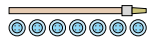


The shoe is                  rubbers long.

Page 1

**Differentiated Challenges***Extra tasks for measuring with blocks and rulers!***Level 1: Foundation (Easier)**

9. How many buttons long is the paintbrush?



The paintbrush is                  buttons long.

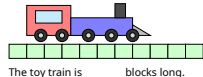
10. Measure the **Glue Stick** with the ruler below.



The glue stick is                  cm.

11. The glue stick is ( **shorter** / **longer** ) than the paintbrush.

12. How many blocks long is the toy train?



The toy train is                  blocks long.

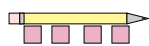
13. Which word is a **unit** of length? Circle the correct answer.

Triangle

Metre

Green

14. Mr Darra measured this pencil with blocks. Did he do it correctly? Circle YES or NO.



YES

NO

Page 3



Mr Darra says: "3 is bigger than 2, so my Glue Stick must be longer!"

**Explain why we cannot measure one thing with rubbers and another thing with pencils to compare their sizes:**

**Reading for Fluency***Let's practice reading like experts!***The 3 Keys to Good Reading**

- Accuracy:** We sound out tricky words instead of guessing.
- Expression:** We don't read like a robot! Our voice goes up at a question mark (?) and we take a breath at a full stop (.).
- Understanding:** We make a movie in our head about what the words mean.

22. **Paired Reading:** Read this story to your partner using the 3 Keys.

Today, Mr Darra showed us how to measure. We learned that we must use the same unit to compare sizes! If Mr Darra uses pencils and Mr Darra uses rubbers, we get confused. Mr Darra raised his hand and asked, "But what if my pencils are short?" Mr Darra smiled. "That is a great question!" he said. Then, Mr Darra asked, "Can we just use a ruler instead?" We all agreed that rulers make measuring fair. Do you know how to use a ruler?

23. **My Reading Checklist:** Tick the boxes after you read!

- ☐ I sounded out the tricky words.
- ☐ I stopped at the full stops.
- ☐ I know what the sentence means.

Page 5

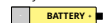
**Part 3: Using a Ruler (cm)**

4. Use the ruler below to measure the marker in centimetres (cm).



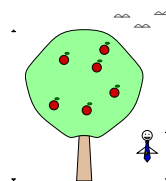
The marker is                  cm long.

5. **Level 2:** This battery does **not** start at 0. Find the length.



Starts at:                  Ends at:                  Length:                  cm.

6. **Tree Challenge:** Measure the **height** of the tree and Mr Darra.



The tree drawing is                  cm tall.

Mr Darra is                  cm tall.

**Part 4: Length Logic**

7. **Ordering (Baseline):** Order these lengths from **shortest** to **longest**.

18 cm,    64 cm,    42 cm,    110 cm

Shortest:                  →                  →                  → Longest:

8. **Estimation:** How tall is Ms                  Circle one.

170 cm                  17 cm                  17 m

Page 2

**Level 2: Intermediate (Medium)**

15. Find the total length if you put the marker and the battery together.

*Hint: Marker = 6 cm, Battery = 3 cm. What is the total?*

Working:                  Total:                  cm.

16. Write a word from your word bank that can be used to measure a curved line.

Answer:

**Level 3: Extension & Reflection**

17. **Super Challenge (Mixed Units):** Order these lengths from shortest to longest.

15 cm,    3 m,    110 cm,    1 m,    80 cm

Shortest:                  →                  →                  → Longest:

18. This stick does **not** start at the first cube. How many cubes long is it?



The stick is                  cubes long.

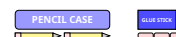
19. Write a sentence explaining **how** you found the length of the stick when it didn't start at the beginning.

20. **Word Problem Challenge:** A piece of wood is 1 metre long. Mr Darra cuts off 20 cm. How long is the wood now?

Working:                  Answer:                  cm

21. **The Measurement Trap!** Mr Darra asked two students to measure items.

- Mr Darra measured a Pencil Case. It was **2 pencils** long.
- Mr Darra measured a Glue Stick. It was **3 rubbers** long.



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24. **Writing: Teach an Alien!**

Write 3 sentences to teach an alien how to measure your desk.

First, I

Next, I

Finally, I

25. **Reflection:** What was the trickiest part about measuring today?

I thought                  was tricky because

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